NACA 2412 Aerofoil

Structural Analysis Project

ANSYS Static Structural + Modal Analysis

Software: ANSYS Workbench 2026 R1 (Student Edition)

Date: April 2026

1. Project Overview

This project is a complete structural analysis of a NACA 2412 aerofoil profile using ANSYS Workbench 2026 R1. The objective was to simulate aerodynamic loading on an aluminium alloy wing section and extract structural performance data — stress distribution, deformation, and natural frequencies.

The project was completed in two phases:

- Day 1 — Static Structural Analysis: Applied a 5000 Pa uniform pressure load on the upper surface to simulate lift, and extracted Von Mises stress and total deformation.
- Day 2 — Modal Analysis: Linked a pre-stressed Modal system to extract the first 6 natural frequencies and mode shapes of the aerofoil.

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2. Geometry Setup — NACA 2412 Profile

2.1 What is the NACA 2412 Aerofoil?

NACA 2412 is a classic aerofoil profile developed by the National Advisory Committee for Aeronautics. The four digits define the shape precisely:

- 2 — Maximum camber is 2% of the chord length
- 4 — Maximum camber is located at 40% of the chord from the leading edge
- 12 — Maximum thickness is 12% of the chord length

This profile is widely used in general aviation (Cessna 172), UAVs, and wind turbine blades due to its excellent lift-to-drag ratio at low speeds.

2.2 Coordinate Data Source

Coordinates were downloaded from the UIUC Airfoil Database (m-selig.ae.illinois.edu). The file uses Selig format — upper surface first (y positive, trailing edge to leading edge), then lower surface (y negative, leading edge to trailing edge). All values are normalised to chord = 1.0.

2.3 Python Script — How It Works

ANSYS SpaceClaim supports Python scripting via its built-in Script Editor. The script performed the following steps:

- Defined all 35 NACA 2412 coordinate pairs as a Python list of (x, y) tuples in normalised form
- Scaled each coordinate by CHORD = 0.5 to convert from normalised units to physical metres
- Built a .NET List [Point2D] using SpaceClaim's Point2D.Create() API — required because SpaceClaim is built on .NET and does not accept native Python lists for typed parameters
- Converted the .NET list to a typed Array using. ToArray()

- Called SketchNurbs.CreateFrom2DPoints (True, pt_array) to generate a closed NURBS aerofoil curve on the active XY plane

The geometry was then manually extruded 12 mm to create a 3D wing section with 0.5 m chord and 12 mm span.

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3. Difficulties Faced and How We Overcame Them

Problem 1 — Script Error: 'expected bool, got list'

The first script version passed a Python list directly to SketchNurbs.CreateFrom2DPoints (). SpaceClaim's .NET-based API rejected this because it expects a typed .NET array, not a Python list.

Solution: Converted the Python list to a .NET List [Point2D] using System.Collections.Generic.List, then called. ToArray() to produce the typed array the API required.

Problem 2 — Script Error: 'expected bool, got Array[Point2D]'

After fixing Problem 1, the arguments were in the wrong order. The SpaceClaim V261 API signature expects the Boolean closed-curve flag as the FIRST argument and the points array as the SECOND.

Solution: Swapped argument order from CreateFrom2DPoints (pt_array, True) to CreateFrom2DPoints (True, pt_array).

Problem 3 — Solver License Limit Exceeded

ANSYS Student Edition is capped at 32,000 nodes and 32,000 elements. With a 1mm mesh on a 500mm chord geometry, the mesh generated over 200,000 elements, triggering the error: 'Your product license has numerical problem size limits.'

Solution: Increased mesh element size from 1mm to 5mm, reducing element count to approximately 15,000–20,000 well within the student licence limit while maintaining sufficient accuracy for the smooth aerofoil geometry.

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4. Material and Boundary Conditions

4.1 Material — Aluminium Alloy

Aluminium Alloy was selected from the ANSYS General Materials library. Key mechanical properties:

Property	Value
Density	2770 kg/m³
Young's Modulus (E)	71 GPa
Poisson's Ratio	0.33
Yield Tensile Strength	280 MPa
Ultimate Tensile Strength	310 MPa

4.2 Boundary Conditions

- Fixed Support: Applied to the root face of the aerofoil. Constrains all 6 degrees of freedom (3 translational + 3 rotational), simulating a wing rigidly attached to a fuselage or test fixture.
- Pressure Load: 5000 Pa applied to the upper surface, directed inward (into the surface). This simulates the net aerodynamic pressure differential responsible for generating lift. A positive pressure value in ANSYS acts inward by default.

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5. What is Von Mises Stress?

Von Mises stress is a scalar value derived from the full 3D stress tensor. It represents the overall intensity of stress at a point and is used in the Von Mises yield criterion to predict when a ductile material will begin to plastically deform.

5.1 The Theory

In 3D, a stress state has six components: three normal stresses and three shear stresses. These are combined into a single equivalent stress:

$$\sigma_{vm} = \sqrt{\left[\tfrac{1}{2} \times \left((\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2 \right) \right]}$$

5.2 The Yield Criterion

- If σ_{vm} < Yield Strength → Material is safe (elastic, no permanent deformation)
- If σ_{vm} ≥ Yield Strength → Material yields (plastic deformation begins)
- If σ_{vm} >> Yield Strength → Risk of fracture

For Aluminium Alloy: Yield Strength = 280 MPa. Any region where Von Mises stress exceeds 280 MPa is yielding.

5.3 Reading the Colour Plot

Blue = minimum stress (near zero), Red = maximum stress. The maximum stress concentration appears at the fixed root near the leading edge this is the classic location for peak bending stress in a cantilever, where the moment arm is greatest.

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6. Static Structural Results

Result	Value	Location
Max Von Mises Stress	5640.5 MPa	Fixed root — leading edge
Min Von Mises Stress	0.063 MPa	Free trailing edge
Average Von Mises Stress	158.99 MPa	Across full body
Max Total Deformation	298.48 mm	Free end — trailing edge tip
Min Total Deformation	0 mm	Fixed root
Average Total Deformation	159.24 mm	Across full body

6.1 Von Mises Stress Plot

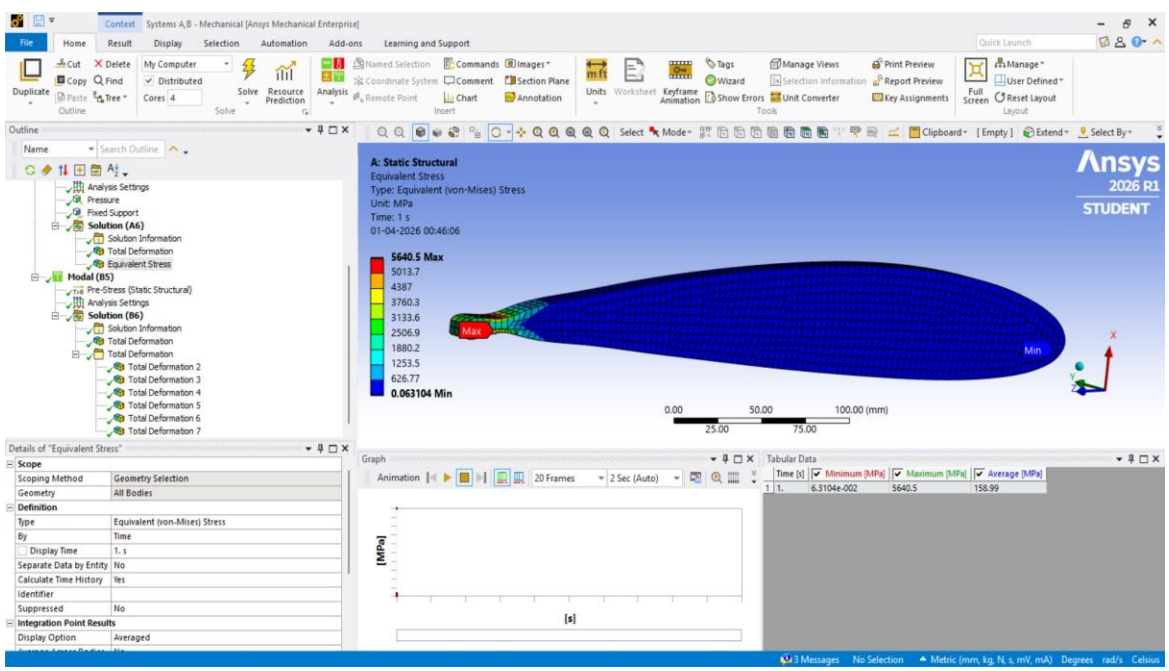

Figure 1: Equivalent Von Mises Stress — Max 5640.5 MPa at fixed root, Min 0.063 MPa at trailing edge

6.2 Total Deformation Plot

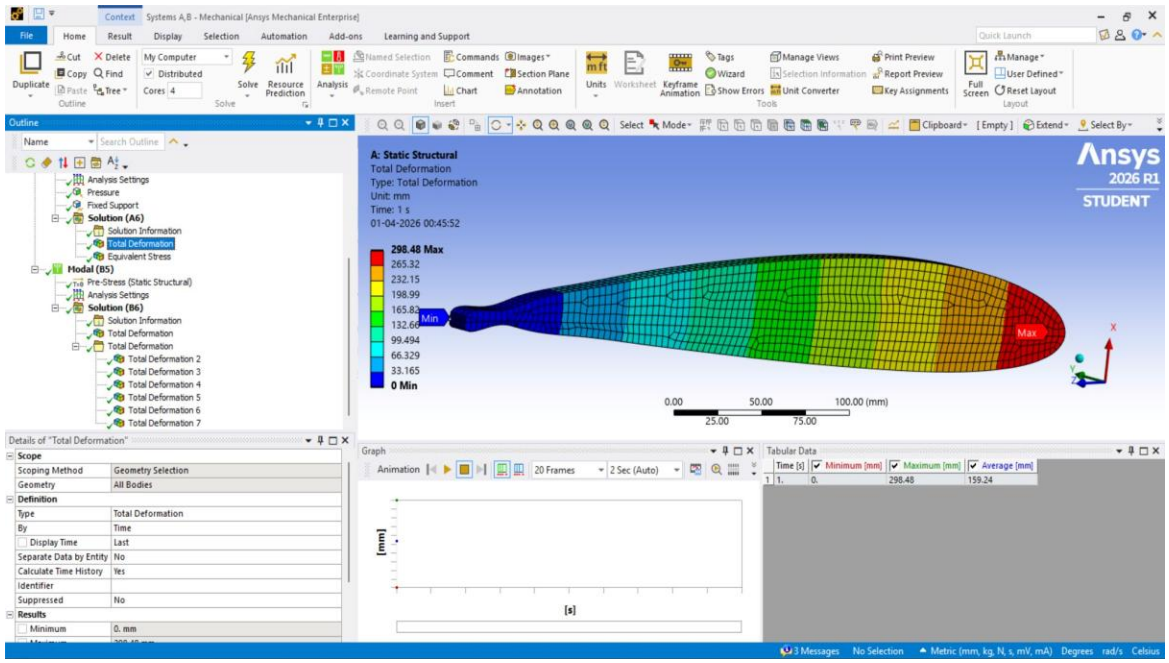

Figure 2: Total Deformation — Max 298.48 mm at free trailing edge tip, 0 mm at fixed root

6.3 Significance of Results

Max Von Mises Stress — 5640.5 MPa

This far exceeds the yield strength of Aluminium Alloy (280 MPa), meaning the material would plastically deform in reality. This is expected given the very thin geometry (12 mm span, 500 mm chord). In a real design, this result would drive a redesign — increasing thickness, adding internal spars, or selecting a higher-strength alloy such as 7075-T6 aluminium.

Max Total Deformation — 298.48 mm

The free end deflects 298.48 mm — approximately 60% of the chord length. This is structurally unacceptable for a real wing but is a direct consequence of the thin extrusion depth. Real aircraft wings are designed to deflect a small fraction of their span even under limit loads. The result correctly demonstrates the cantilever bending behaviour expected for a fixed-free beam.

Average Stress — 158.99 MPa

The average stress is below the yield strength, confirming that while stress concentrations at the root are extreme, the majority of the structure is within the elastic regime. This is characteristic of bending-dominated problems where stress varies linearly from maximum at the root to zero at the free tip.

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7. Modal Analysis

7.1 What is Modal Analysis?

Modal analysis determines the natural frequencies and mode shapes of a structure the specific frequencies at which it will resonate if disturbed by an external force. Every structure has an infinite number of natural frequencies but the first few (lowest) are most critical because they are easiest to excite with real-world forces.

If an external force matches a natural frequency, resonance occurs the structure oscillates with amplitudes that grow rapidly, potentially causing catastrophic failure even from small input forces. This is why modal analysis is mandatory in aerospace, automotive, and civil engineering.

7.2 Pre-Stressed Modal Analysis

This analysis was run as a pre-stressed modal analysis the Modal system was linked to the Static Structural solution in ANSYS Workbench. This means the natural frequencies were calculated accounting for the stress stiffening effect of the 5000 Pa pressure load. A pre-stressed modal analysis is more accurate than an uncoupled modal analysis for structures under significant load.

7.3 Natural Frequency Results

Mode	Frequency (Hz)	Description
1	21.792 Hz	1st bending — out-of-plane, trailing edge sweeps up/down
2	22.798 Hz	2nd bending — similar to Mode 1, slightly higher frequency
3	221.77 Hz	3rd bending — higher order spanwise bending
4	317.29 Hz	Torsional / combined bending-torsion mode
5	585.75 Hz	Higher order bending with internal node
6	686.11 Hz	Complex combined mode — multiple nodes visible

7.4 Mode Shape Plots

Mode 1 — 21.792 Hz (1st Bending Mode)

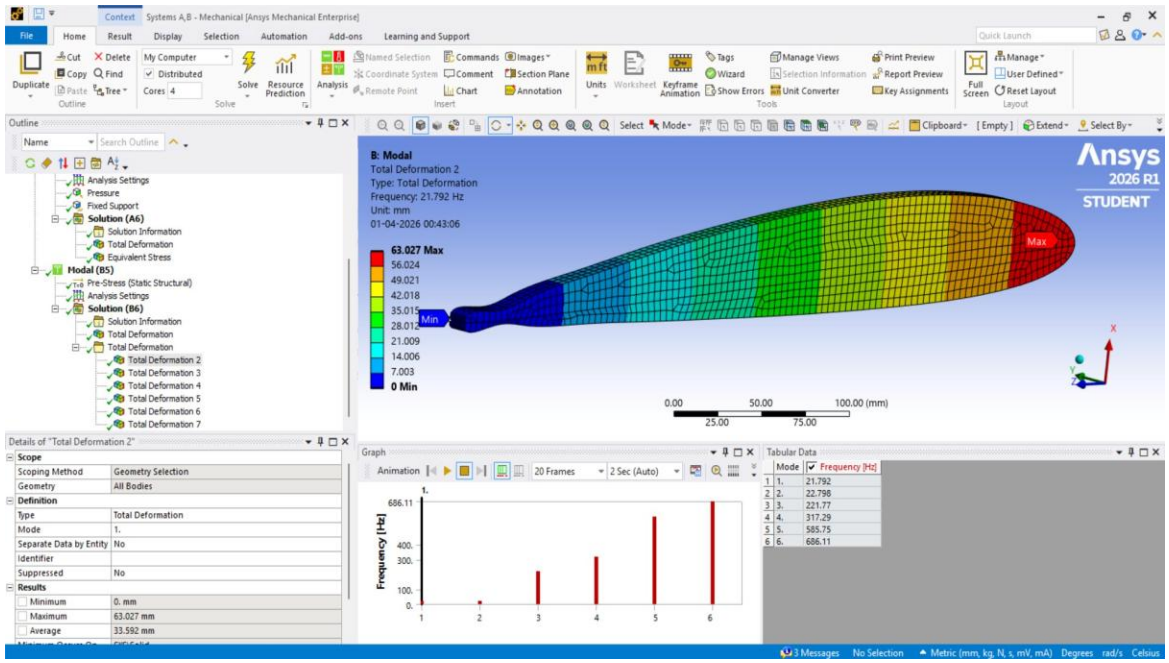

Figure 3: Mode 1 — 21.792 Hz | Max deformation 63.027 mm at trailing edge tip

Mode 2 — 22.798 Hz (2nd Bending Mode)

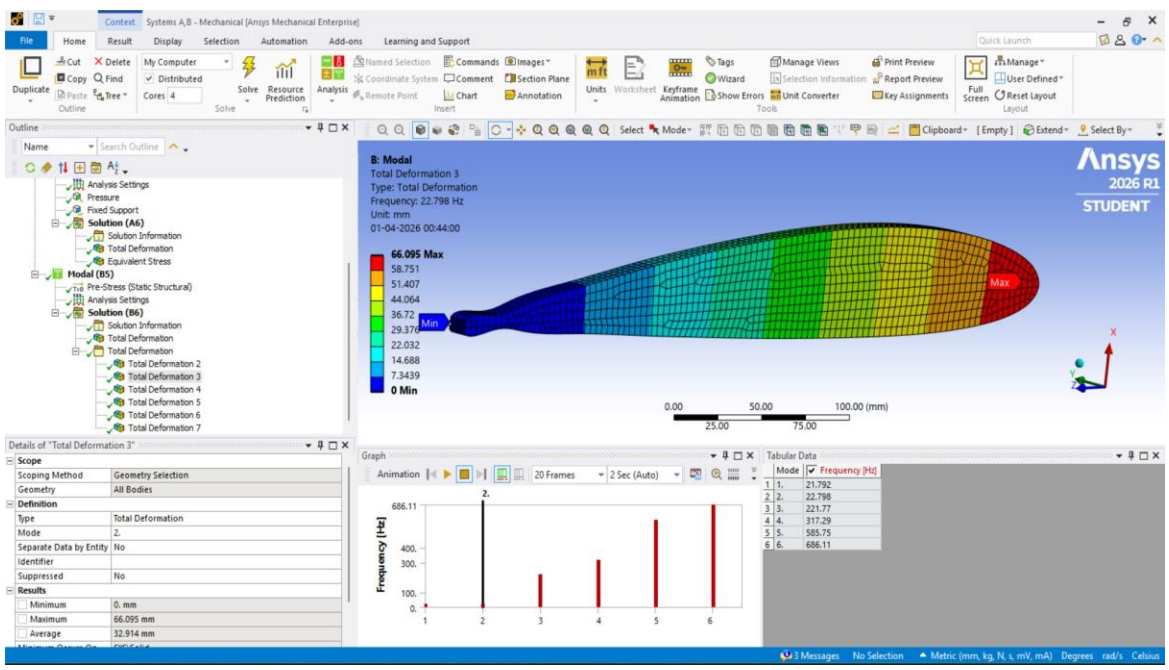

Figure 4: Mode 2 — 22.798 Hz | Max deformation 66.095 mm

Mode 3 — 221.77 Hz (3rd Bending Mode)

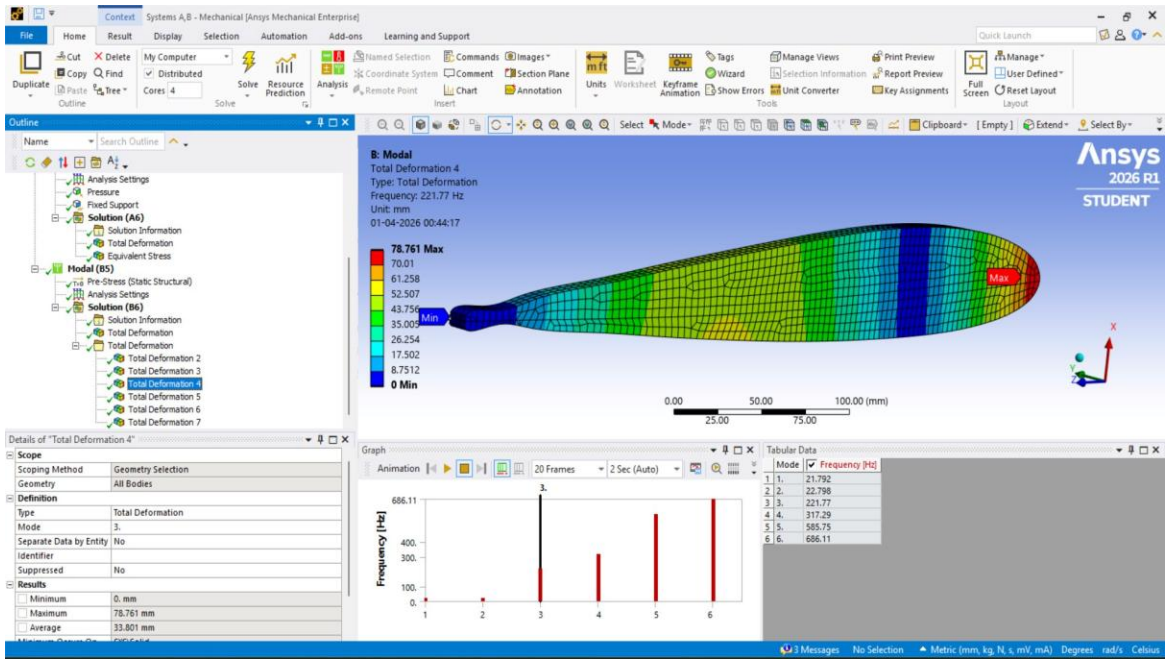

Figure 5: Mode 3 — 221.77 Hz | Max deformation 78.761 mm | Higher order spanwise bending

Mode 4 — 317.29 Hz (Torsional Mode)

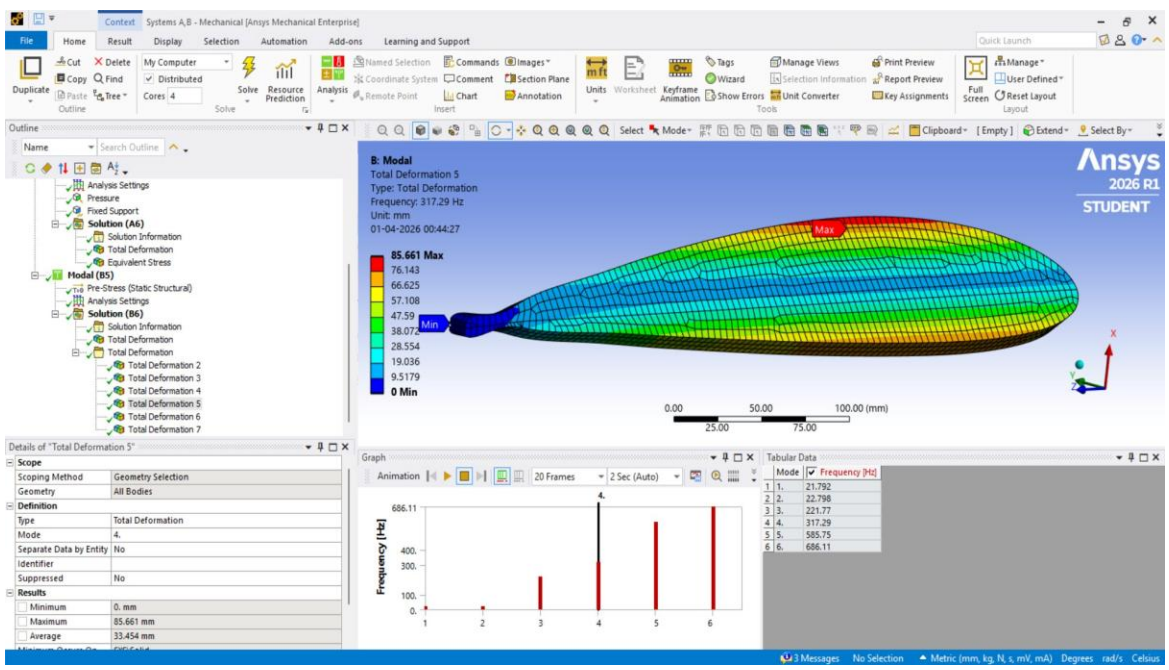

Figure 6: Mode 4 — 317.29 Hz | Max deformation 85.661 mm | Torsional/bending combined

Mode 5 — 585.75 Hz

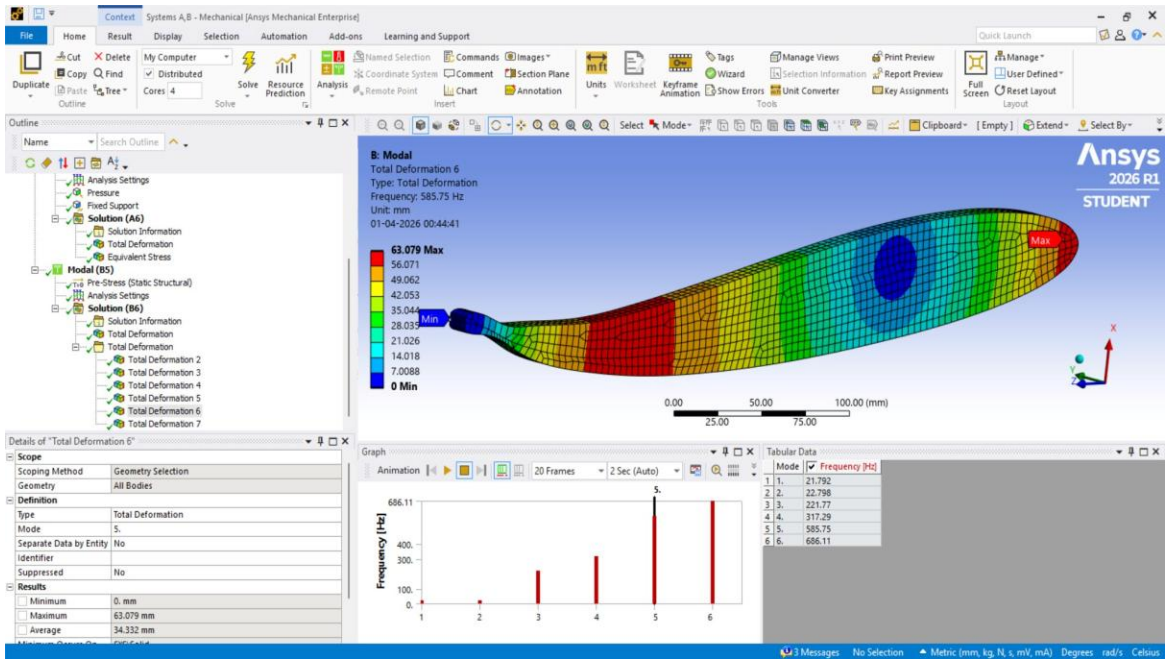

Figure 7: Mode 5 — 585.75 Hz | Max deformation 63.079 mm | Internal node visible in deformation pattern

Mode 6 — 686.11 Hz

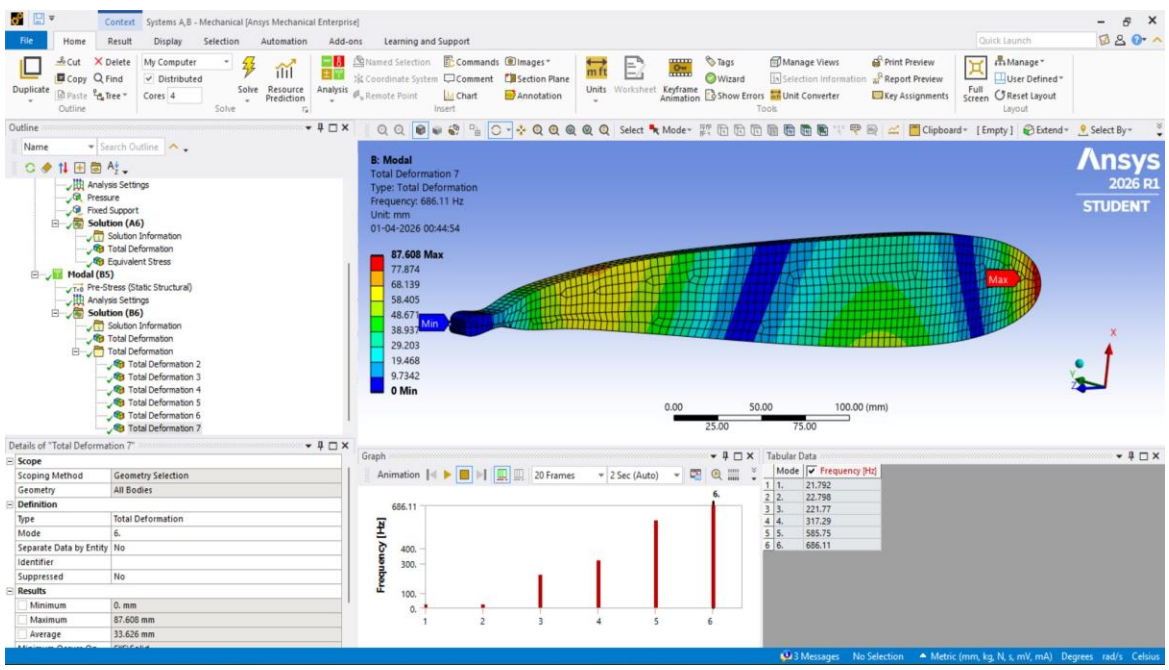

Figure 8: Mode 6 — 686.11 Hz | Max deformation 87.608 mm | Complex multi-node mode shape

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8. Real World Applications

8.1 Aircraft Wing Design

Every commercial and military aircraft wing undergoes FEA (Finite Element Analysis) before manufacture. Von Mises stress analysis confirms no yielding under limit loads, while modal analysis ensures wing natural frequencies are safely separated from engine excitation frequencies to prevent resonance and flutter.

8.2 Aeroelastic Flutter Prevention

Wing flutter is a dangerous phenomenon where aerodynamic forces couple with structural natural frequencies, causing self-sustaining oscillations that can destroy a wing in seconds. The flutter speed must be well above the maximum operating speed. Modal analysis provides the natural frequency data needed to predict and prevent flutter.

8.3 Wind Turbine Blades

Wind turbine blades use aerofoil cross-sections along their span. Modal analysis ensures blade natural frequencies do not coincide with the rotor's operating frequency harmonics (1P, 3P) which would cause resonance fatigue and blade failure.

8.4 Structural Health Monitoring

Natural frequencies serve as a structural fingerprint. In service, a downward shift in measured frequencies indicates damage (cracks, delamination) has reduced structural stiffness. This forms the basis of non-destructive structural health monitoring used on aircraft, bridges, and offshore structures.

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9. Complete Project Parameters Summary

Parameter	Value
Aerofoil Profile	NACA 2412
Chord Length	500 mm (0.5 m)
Extrusion Span	12 mm
Material	Aluminium Alloy (ANSYS library)
Young's Modulus	71 GPa
Yield Strength	280 MPa
Boundary Condition	Fixed Support at root face
Load	5000 Pa uniform pressure on upper surface
Mesh Element Size	5 mm

Parameter	Value
Max Von Mises Stress	5640.5 MPa
Max Total Deformation	298.48 mm
Mode 1 Frequency	21.792 Hz
Mode 2 Frequency	22.798 Hz
Mode 3 Frequency	221.77 Hz
Mode 4 Frequency	317.29 Hz
Mode 5 Frequency	585.75 Hz
Mode 6 Frequency	686.11 Hz
Software	ANSYS Workbench 2026 R1 Student